
Slide 1: Title Slide

"Good morning everyone. My name is Lin Zhi-Long, a Master's student from the AI department at Taiwan Tech. Today, I am presenting a review on 'State-of-Health Estimation for Lithium-ion Battery Packs'. This work is based on a recent paper published in the *Journal of Energy Storage*. Let's get started."

Slide 2: Outline

"Here is the outline for my presentation. I will start with an introduction to why this topic matters. Then, we will look at how batteries degrade, how we extract health features, and the different models used to estimate their health. Finally, I will discuss the current challenges and future trends before concluding ²."

Slide 3: Introduction

"First, why is State of Health, or SOH, so important? It is the cornerstone of safety for electric vehicles.

However, most research only looks at single battery cells. But in the real world, EVs use Battery Packs, which are made of many cells connected together.

The problem with packs is the 'Barrel Effect'. Just like a wooden barrel can only hold water up to its shortest plank, a battery pack's performance is limited by its weakest cell. This Incompatibility between cells makes packs age faster than single cells, and this is the research gap we are addressing today."

Slide 4: How Lithium-ion Batteries Work

"To understand degradation, we first need to know how the battery works. Please look at Figure 1.

It shows the core mechanism. Ideally, lithium ions move back and forward between the positive and negative electrodes through the separator.

When charging, ions move to the anode, and when discharging, they move back to the cathode. It is this movement that stores and releases energy."

Slide 5: Cell-Level Degradation Mechanisms

"Unfortunately, this process isn't perfect. Figure 2 here summarizes the degradation mechanisms inside a cell.

Chemically, we have the growth of the SEI film, which is a layer that consumes active lithium and increases resistance. We also have electrolyte decomposition.

Mechanically, the particles inside the battery expand and contract during use, causing stress and structural damage over time."

Slide 6: Impact of Fast Charging

"Fast charging makes these problems worse.

High currents create Thermal Stress, raising the temperature and speeding up chemical reactions.

A major risk is Lithium Plating. Instead of going inside the anode, lithium builds up on the surface, forming 'dendrites' which can cause short circuits. This is especially dangerous in cold weather"

Slide 7: Degradation Mechanism of Battery Packs

"Now, let's move to the pack level, which is our main focus.

Please look at Figure 3 on the below. This simulation the 'Barrel Effect' that I mentioned earlier. The weakest cell restricts the whole pack that you can see in the Fig 3.

Figure 4 shows the 'Vicious Cycle' of Incompatibility. Small manufacturing differences lead to uneven heat and current. This causes some cells to age faster than others and increasing the Incompatibility even more. It is a loop that accelerates failure ."

Slide 8: Health Feature Extraction - ICA

"So, how do we measure this health?

One effective method is Incremental Capacity Analysis, or ICA.

If you look at Figure 5, you can see the curves change as the battery ages. The peaks shift position and drop in height. The peaks tell us the unique status of the battery's health.

However, this method is very sensitive to noise, so we need good data smoothing."

Slide 9: Health Feature Extraction - Time & Frequency

"We can also use Time and Frequency domains.

For Time-based features, we simply measure how long it takes to charge. As batteries age, the constant voltage charging time gets longer. It's simple and fast.

For Frequency domain, shown in Figure 6, we use Impedance Spectroscopy or EIS. It gives deep insight into internal chemistry, but the equipment is expensive and slow."

Slide 10: Feature Comparison Table

"Let's look at **Table 1** now. This table compares all methods to find battery health features. We see three main trade-offs.

First, **ICA and Time-based** methods are very practical because they only need simple voltage and current data. However, they are easily affected by noise and need strict charging rules.

Second, **Frequency Domain (EIS)** gives rich, sensitive data. But the cost is very high, and measuring takes too much time.

Finally, the most important one for packs is **Inconsistency**. This method directly finds the pack's degradation problem. But the challenge is that it is difficult to check this data in real-time online. We need to fix this difficulty.

Slide 11: SOH Estimation Models - Statistical Filters

"we have the features, we need a model to calculate SOH.

First, we have Statistical Filters like the Kalman Filter.

Think of this as a 'White Box' approach. It uses math equations to estimate the state.

The advantages is that it is fast and good for real-time use.

The disadvantages is that it relies heavily on having a perfect starting model"

Slide 12: SOH Estimation Models - Data-Driven

"Next is the **Data-Driven Approach**, or the 'Black Box' method. As you see in **Figure 8**, we use the Neural Network to let the computer learn the SOH directly from the large amount of data. This works well for complex, non-linear problems.

We have five main types here. **SVM** is simple and good for smaller data sets. **RVM** and **GPR** are great because they give us the prediction and also the **confidence level** and telling us how reliable the result is. **NN** handles the most complex tasks, and **ELM** is the fastest method to train. These data-driven models are very precise.

Slide 13: SOH Estimation Models - Hybrid

"Finally, we have Hybrid Methods³⁴.

This combines the best of both worlds—for example, using a Kalman Filter together with a Neural Network.

This offers the highest accuracy and robustness, but as you can guess, it is computationally expensive to run."

Slide 14: Comparison of Models (Qualitative)

"Here is a qualitative comparison.

Filters are the 'Fast' option, great for real-time chips.

Machine Learning is the 'Precise' option, but data-hungry.

Hybrid Methods are the 'Balanced' option, offering the best performance if your system can handle the complexity ."

Slide 15: Table 2 Comparison

Let's look at **Table 2** now. We see a clear choice: speed versus accuracy. **Statistical Filters** are the fastest choice for real-time systems. But they are not precise and require a perfect model.

Machine Learning models are very accurate, and they handle complex data well. But they need high computing power, making them complex. This is why **Hybrid Methods** are the future. They offer the best overall accuracy, balancing speed and performance.

Slide 16: Estimation Error Comparison

"Now, let's look at the actual numbers.

Filter-based models typically have an error rate around 1.2 to 3.37%.

Machine Learning models, like SVM, can push this error down to below 0.5%.

And advanced Hybrid methods have achieved errors as low as 0.014 in some studies. This shows a clear trend toward higher accuracy with more advanced models."

Slide 17 & 18: Table 3 Detailed Synopsis

"Now we look at the numbers in **Table 3**. This table shows how the estimation error goes down as we use more advanced models. Early **Statistical Filters** often have a higher error, sometimes up to **3.37%**. Then, simple models like **SVM** quickly reduce this error to under

0.5%. This shows the high precision of machine learning. **ELM** also gives very low errors, around **0.08%**. **GPR** is important because it can estimate health using only **partial charging data**. Finally, **Hybrid Methods** give us the best performance. They achieve the lowest error, reported as low as **0.014**. This confirms that combining models gives the highest accuracy."

Slide 19: Challenges

We now move to **Slide 19**, discussing the current challenges we face. The biggest problem is the '**Lab-to-Reality**' Gap. Look at the second point: our models are trained with very clean data in the lab. But in the real world, the battery data is **noisy** and has many sensor errors. Also, **Environmental Adaptability** is a big issue. When driving, the temperature changes often, and this makes our estimation methods fail because the model expects a stable environment.

Slide 20: Future Prospects

Let's now look at the solutions and our **Future Prospects** on **Slide 20**."

"Looking to the future, we have three main goals. First, we must adapt our models for new battery types, like **Solid-State Batteries**. Second, we will use **Advanced Sensors** such as **Ultrasonic** to help us see inside the battery without destroying it. Our most important goal is to put the accurate **Hybrid Models** directly into the car's **BMS**. This will allow the system to manage the battery smartly, making it much safer and last longer."

Slide 21: Conclusion

"In conclusion, we must focus on **Battery Packs** because **Cell Inconsistency** is the main problem that speeds up degradation. The best solution is the **Hybrid Model**, which gives us high accuracy while being efficient. Our final goal is clear: we must integrate these robust models into the real-time **BMS** to ensure safer and more reliable operation in the future."